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2023

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Sections as directed.

Section – A

Answer any **four** questions of the following : 10×4 = 40

1. What is computer network ? List the properties of a good network. What are the advantages and disadvantages of computer network ?

2. Discuss TCP/IP model. Describe and explain the layers of TCP/IP Model. What are the merits and demerits of TCP/IP model ?

3. What is network topology ? Describe the functions, advantages and disadvantages of ring topology, bus topology and tree topology.

4. What do you mean by transmission media ? Explain guided and unguided media and discuss their types.

5. What is transport layer ? Write the functions of transport layer. Discuss connection management the three events involved in the connection.

6. What is the format of IPv4 header ? Describe the significance of each field.

7. Assume Data frame is 1101011011 and generator polynomial, G(x) is x + x + 1. Calculate the transmitted frame using CRC method.

8. Write notes the following :

   (a) SNMP

   (b) FTP

   (c) Process to process delivery

Section – B

9. Answer **all** questions of the following : 3×10 = 30

(a) What are the functions of session layer in OSI model ?

(b) What is Routing ?

(c) What is MAC address ?

(d) What are the repeaters in computer network ?

(e) What are the challenges of computer network ?

(f) Differentiate between serial and parallel transmission.

(g) Describe distortion and noise in signal Transmission.

(h) What is port number ?

(i) Compare TCP and UDP protocols.

(j) What is Parity bit ?

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2022(New)

Time : 3 hours

Full Marks : 70

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The figures in the margin indicate full marks.

Answer from both the Groups as directed.

Group – A

Answer any **four** questions of the following : 10×4 = 40

1. Differentiate OSI reference model with the TCP/IP reference model.
2. Explain the different topologies of the network.
3. Explain the parity bit method used for error detection. Suppose a bit sequence 1111101 is received. Assume even parity bit method is used, find whether it has been received correctly or not. if not, correct it.

4. What are the design issues of Data Link Layer ? Explain.

5. What is the need of Flow control ? Explain the common approaches for flow control in data link layer.

6. Explain the services of transport layer.

7. Explain the significance of switching. What are the different switching techniques used in computer networks ? Discuss.

8. What is IP address class ? Describe various classes in IP address class.

Group – B

Answer all questions of the following : 3×10 = 30

9. What is the role of protocols in network ? Explain HTTP.

10. What is the difference between Broadcasting and Multicasting ?

11. What are the characteristics of datagram network ?

12. Define congestion in computer networks.

13 Explain about the Radio Transmission.

14. What are the disadvantages of netwrorking ?

15. What is port address ?

16. What are the pros and cons of wireless communication system ?

17. Name any three network devices and their uses.

18. What are the layers of TCP/IP model ?

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New

2021

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Groups as directed.

Group – A

Answer any **four** questions of the following : 10×4 = 40

1. Write a detailed note on TCP / IP reference Model. Also, compare TCP / IP and OSI reference model.

2. Explain the function of TCP / IP protocol.

3. Describe the congestion control algorithms.

4. Explain the design issues of datalink layers.

5. Describe the structure of network layer in the internet.

6. Discuss the services offered by the application layer.

7. Explain network architecture in detail.

8. Explain the bus topology and ring topology networks. Compare their performance.

Group – B

Answer **all** questiions : 3×10 = 30

9. Write difference between Hub and Switch ?

10. Explain transmission impairments.

11. What are the types of LAN technologies ?

12. What are the functions of Router ?

13. Discuss the use of computer networks.

14. Compare TCP and IP services.

15. What are the objectives of computer communication networks ?

16. What is the significance of flow control ?

17. Define Error detection and correction.

18. What are the main features of Internet Architecture ?

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New

2020

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Groups as directed.

Group – A

Answer any **four** questions : 10×4 = 40

1. What are the various types of networks ? Differentiate between LAN, MAN and WAN in terms of topology and technology used by each network.

2. What do you mean by flow control in data link layer ? Discuss with diagrams the working of Selective Repeat Method and compare it with GO Back N.

3. Which layer ensures process to process or end to end delivery of an entire message ? Explain TCP Connection and describe use of each field in TCP header.

4. (a) Explain the importance of application layer.

   (b) Write short notes on the following :

   (i) SMTP

   (ii) SNMP

5. Explain the need of layering in data communication protocol stack. List and explain briefly each layer of OSI model.

6. Explain the TCP/IP model in detail. What are the various layers of TCP/IP model and list their functions ?

7. What are the functions of Network layer ? Explain Message switching, packet switching and circuit switching with proper diagrams.

8. How many types of error correction are there in data. link layer ? Describe Hamming error detection and correction by taking suitable examples.

Group – B

Answer **all** questions : $3 \times 10 = 30$

9. List guided transmission mediums.

10. What is the difference between broadcasting and multicasting ?

11. What do you mean by parity bit ?

12. Explain the role of repeaters in analog and digital system.

13. Differentiate between virtual circuit and diagram.

14. What is the difference between MAC address and IP address ?

15. What are transmission impairments ?

B-54

16. Why twisted pair cable is twisted ?

17. What is signficance of protocols ?

18. Differentiate between Radio Communication and Satellite Communication ?